
EDUCATION

University of Ontario Institute of Technology (Ontario Tech University)

Oshawa, Ontario

BSc, Computer Science

September 2022 – May 2027 (expected)

SKILLS

Programming & Development: Java, Python, C++, C, HTML, CSS, JavaScript, RTOS, Bazel, CI/CD, WSL, EC2

Data & Mathematical Computing: NumPy, pandas, Matplotlib, Logic Analyzer.

Embedded Systems: STM32, ESP32, Raspberry Pi, Arduino, Linux, CANDB++, BUS Master.

Software: MATLAB, Simulink, Git Lab, GitHub, Arduino IDE, Cube IDE, CubeMX, CANoe, Jira

WORK EXPERIENCE

Rivian and Volkswagen Group Technologies

Vancouver, BC

Software/Systems Engineer Intern

May 2026- Present

- Built a Python-based agent using the Claude SDK to convert systems-engineering requirements (Jama) into pytest test scripts, enabling test-case generation before firmware implementation.
- Synchronized requirements and test cases bidirectionally, maintaining traceability as requirements changed throughout development.
- Used Bazel to build .so targets from embedded C source and validated them via Python/pytest-based SIL test scripts against system requirements.
- Triaged SIL test failures and root-caused defects with embedded software engineers to drive resolution.
- Partnered with test automation engineers to enforce coding standards and consistency across agent-generated test scripts.

Ontario Tech University ACE (Automotive center of excellence)

Oshawa, ON

Engineering Student

May 2025– May 2026

- Designed a diagnostic controller to validate temperature and humidity (Modbus TCP/IP and MQTT).
- Developed a Python + Flask API based real-time web dashboard for monitoring, configuration, and control.
- Prepared detailed software validation test plans for threshold logic, alert triggers, and system reliability.

DESIGN TEAMS

Ontario Tech Racing (Formula SAE Team)

Oshawa, ON

Embedded Systems Lead

Sept 2024 – Present

- Lead a 6-member embedded team, define coding and documentation standards, manage development workflows.
- Developed integrated and validated ECU firmware in Simulink (Stateflow + Embedded Coder), implemented throttle/brake mapping and motor control with custom DBCs, and shutdown-circuit safety logic.
- Programmed BMS firmware for Analog Devices LTC6813-1, utilizing isoSPI, cell voltage and temperature monitoring, passive cell balancing.
- Designed ECU and BMS models (Simulink) for virtual testing, SIL validation, and logic error discovery. Performed automated CANoe and pytest-based SIL test scripts and bench-level HIL setups for motor controller validation.

Aerecon Drone Design Team

Oshawa, ON

Software Lead

Nov 2024 – Present

- Implemented PID-based flight stabilization for brushless motors using PWM-driven ESCs.
- Developed Python visualization tools IMU data and integrated GPS for NMEA-based position tracking.

PROJECTS

FSAE Car Telemetry ([link](#))

- Built a web interface using Async Web Server and REST API to monitor vehicle data remotely. Built and integrated on-car (ESP32 + HMI) and cloud (Firebase + TypeScript) telemetry systems, collaborating with Vehicle Dynamics and HV teams and supporting FSAE Michigan technical inspections.